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From Randomness to Predictability - Fine Tuning the Key Parameters of LLM for Better Detection of Hallucination

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***Abstract*—Large Language Models (LLMs) are revolutionizing various fields but remain vulnerable to hallucinations, undermining trust and reliability. In this paper, we have introduced a novel mathematical model that can predict the level of hallucination in various architectures of LLMs. This new model takes into account the key parameters and architectural factors of different models of LLMs to arrive at its prediction. The obtained solution is proven to be positive and can predict hallucination within the range of 10% error margin in various LLM models. To validate the theoretical results, we conducted numerous experiments which provided insights into the behavior of the model and under different set ups. This new finding aims to be a stepping stone to predict hallucination in LLM very effectively.**

***Index Terms*—LLM, HaluEval, Top P, Top K, Temperature, Context Window, Number of Tokens, Number of Layers, Embedding Dimension, Attention Heads.**

I. Introduction

In the current era rapid advancements in Large Language Models [1] have major applications in all domains of human life like medicine, education, legal and automation in customer service [2]. One of the most important shortcomings of LLM is Hallucinations. Hallucinations by definition means deviation from facts and potential logic [3] [4] [5] They may range from minor inconsistencies to major deviations that are completely fabricated. To comprehensively understand the hallucination, three main areas that should be addressed are: Identifying source, Detection and Mitigation [6]. Majority of the research that has been conducted is in understanding the underlying causes of hallucination, rather than detecting and mitigating it. This study aims to bridge this research gap by conducting an empirical study focused on detecting factual hallucinations, one of the most critical error sources in LLMs and thereby reducing randomness and increasing predictability.

A. Exploration of Key Parameters and Architectural Factors

The creative outputs generated by LLMs are controlled by key parameters and architecture of the model that influence how these models generate text. This deeper exploration focuses on five critical parameters

1) Top P Sampling (Nucleus Sampling): Nucleus Sampling or Top P sampling is more adaptive and is a parameter that utilizes cumulative probability to dynamically adjust the number of tokens [7]. Lower values constraint token selection to the most probable outcomes, while higher values increase creativity and diversity.This study takes Top P values ranging from 0.01-1.

2) Top K Sampling: When LLMs try to generate the next text, one of the major methods to constraint the number of words or characters which are also called tokens is Top K sampling [8]. It has been found that a large value of K leads to more hallucination. In this study the Top K values range from 1-100.

3) Temperature: One of the main parameters that scales the log probabilities for adjusting the randomness is Temperature [9]. When the temperature is high, hallucination is high. In this study, we have taken the temperature values ranging from 0.1 to 2.

4) Context Window: Context Window size determines the amount of previous text that can be utilized for the next text generation. Thus, a logically larger context window implies more context thereby improving coherence and vice versa [10]. The range of Context Window in this study is from 1- 600.

5) Number of Tokens: Number of tokens determines the maximum number of tokens that are present in the response. Focused answers are obtained by lower max-tokens and vice versa [11]. The number of tokens lie within 1 to 1024 in this study.

Fine-tuning the above five parameters have proven to enhance the accuracy of hallucination detection. In this research, we experimented with various combinations of these parameters to predict the levels of hallucination more effectively.

B. Key LLM Model Types in LLMs

The Transformer architecture is widely used for LLMs due to its parallelizability and capacity, enabling the scaling of language models to billions or even trillions of parameters. Existing LLMs can be broadly classified into three types: encoder-decoder, causal decoder, and prefix decoder.

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1) Encoder-Only Models: Focus on understanding text (e.g., BERT) for tasks like text classification and sentiment analysis by generating contextual embeddings [12].

2) Decoder-Only Models: Specialize in text generation (e.g., GPT) for tasks like machine translation and dialogue generation [13].

3) Encoder-Decoder Models: Combine both components (e.g., T5) for tasks requiring input-output sequence mapping, such as translation and summarization [14].

TABLE I
LLM MODELS USED IN THIS STUDY

Model Type	Key Purpose	Notable Models	Research Use Cases
Encoder	Feature extraction, contextual embeddings	BERT	Text classification
Decoder	Autoregressive generation	GPT	Text generation
Encoder-Decoder	Complex transformations	T5	Machine translation, summarization

C. Architectural aspects influencing Hallucination

In addition to fine-tuning generation parameters, architectural factors such as the number of layers, attention heads, and embedding dimensions play a significant role in hallucination dynamics.

1) Number of Layers: Deeper models capture complex semantic patterns but are more prone to reinforcing hallucinations by amplifying early-stage errors [15].

2) Attention Heads: Multiple attention heads enhance contextual understanding but may overfit to noise, leading to hallucinated outputs [16].

3) Embedding Dimensions: Higher embedding dimensions provide richer feature representations but can also encourage spurious correlations, increasing hallucination risk [17].

TABLE II
ARCHITECTURAL PARAMETERS OF VARIOUS MODELS.

Model	Number of Layers	Embedding Dimension	Attention Heads
BERT-Large	24	1024	16
XLNet-Large	24	1024	16
GPT-3 (175B)	96	12288	96
PaLM 8B	32	4096	32
mT5-Base	12	512	12
T5-Large	24	1024	16

Developing a mathematical model to predict the level of hallucination in LLMs enhances the reliability and accuracy of LLMs in various applications. The novelty of the proposed mathematical model has the following points to it

1) Takes into account the key parameters
2) Factors in the various types of architectural models that are most commonly used
3) Includes the important architectural parameters of various models

This model gives a framework for predicting the level of hallucination by adjusting the various parameters of the architectures.

D. Model Formulation

For this research we have used the HaluEval Question and Answer dataset. Within the overall dataset, there is a QA dataset of 10,000 questions with answers which contains both correct and hallucinated answers[16].

II. METHOD

A. Development of Parameter-based Weighted Formula

1) Initial Framework : We developed a weighted formula approach based on the hallucination evaluation framework (HalluEval). The base architecture was modified to analyze the relationship between the 5 parameters and hallucination rates using the GPT2 model.

2) Statistical Analysis: For each parameter, we conducted polynomial regression analysis to determine the optimal relationship between the parameter and hallucination rate [17]. The selection of polynomial degree was based on:

1) R-squared values which will be used as weights
2) Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) for finding the best fit model [18].

The analysis revealed the following relationships:

Top P Function $f(P)$:

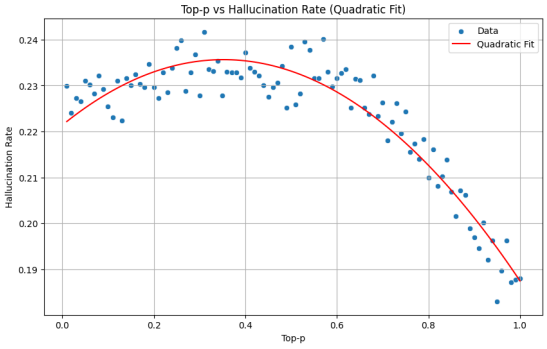


Fig. 1. Top P vs Hallucination rate

$$E_{\text{Top } P} = 0.2214 + 0.0810p - 0.1150p^2 \quad (1)$$

$$R^2_{\text{Top } P} = 0.904 \quad (2)$$

Top K Function $f(K)$:

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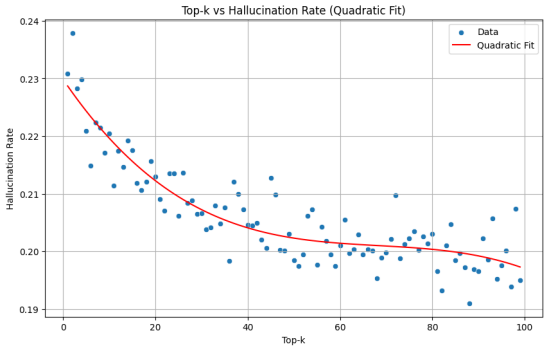


Fig. 2. Top K vs Hallucination rate

$$E_{\text{Top } K} = 0.2299 - 0.0012k + 1.664 \times 10^{-5}k^2 - 8.121 \times 10^{-8}k^3 \quad (3)$$

$$R^2_{\text{Top } K} = 0.812 \quad (4)$$

Temperature Function $f(T)$:

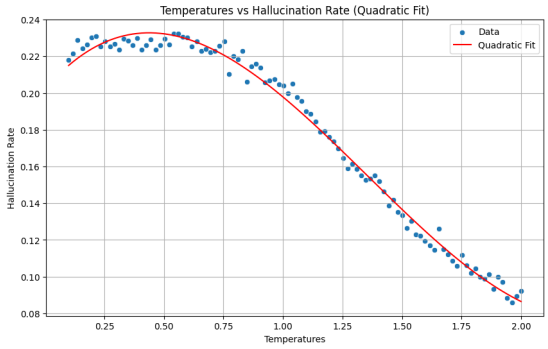


Fig. 3. Temperature vs Hallucination rate

$$E_{\text{Temp}} = 0.2020 + 0.1505t - 0.2051t^2 + 0.0505t^3 \quad (5)$$

$$R^2_{\text{Temperature}} = 0.992 \quad (6)$$

Context Window Function $f(C)$:

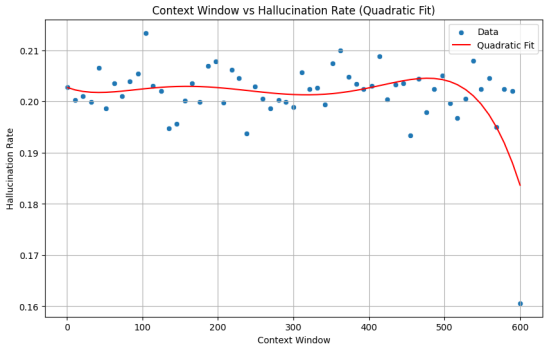


Fig. 4. Context Window vs Hallucination rate

$$E_{\text{Context Window}} = 0.2028 - 5.589 \times 10^{-5}c + 9.832 \times 10^{-7}c^2 \quad (7)$$

$$R^2_{\text{Context Window}} = 0.292 \quad (8)$$

Number of Tokens Function $f(N)$:

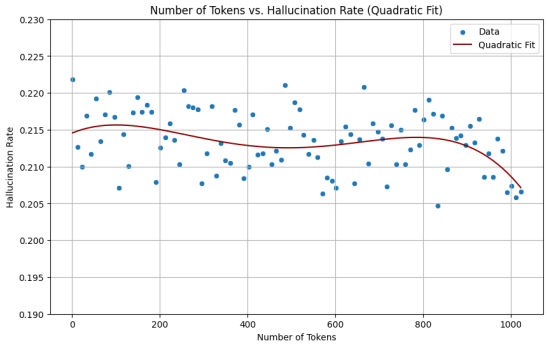


Fig. 5. Token Count vs Hallucination rate

$$E_{\text{Number of Tokens}} = 0.2146 + 2.446 \times 10^{-5}n - 1.622 \times 10^{-7}n^2 \quad (9)$$

$$R^2_{\text{Top Token Count}} = 0.159 \quad (10)$$

Inference: From the above Figures 1-5, derived polynomial functions indicate that each parameter independently influences hallucination probability.

3) Weight Assignment: Based on the statistical significance and R^2 values, we assigned normalized weights to each parameter [19]:

1) Temperature: 0.30
2) Top-p: 0.25

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3) Top-k: 0.20
4) Context window: 0.15
5) Number of tokens: 0.10

4) Weighted Metric: The key parameters individually contribute to hallucination along with their interplay with other factors. To represent the contribution of both these values, we came up with a weighted metric (W) taking all the above values into consideration.

$$W = (E_{Temp} \times 0.30) + (E_{Top_p} \times 0.25) + (E_{Top_k} \times 0.20) + (E_{Context_Window} \times 0.15) + (E_{Number_of_Tokens} \times 0.10) \quad (11)$$

B. Architectural Impact Analysis

1) Layer Dimension Factor: We established a logarithmic relationship between layer dimensions and hallucination rates:

$$f_{layer} = -0.05 \log\left(\frac{d_{layer}}{10000}\right) \quad (12)$$

The natural logarithm relationship was chosen for layer dimensions as it exhibited diminishing returns in hallucination reduction as layer sizes increase [20]. The 10,000 normalization factor was chosen based on typical baseline dimensions in large language models.

2) Attention Head Factor: A square root relationship was determined for attention heads:

$$f_{head} = -0.05\sqrt{\left(\frac{n_{head}}{12}\right)} \quad (13)$$

The relationship captures the diminishing returns observed when increasing the number of attention heads. The square root relationship for attention heads was established as the square root function captures this nonlinear improvement pattern[21]. By empirical analysis, the division by 12 normalizes the value relative to common transformer architectures.

3) Model Depth Factor: A linear relationship was established for the number of layers:

$$f_{depth} = 0.02\left(\frac{n_{layers}}{12}\right) \quad (14)$$

This reflects the observed linear correlation between model depth and hallucination tendencies. The linear relationship for the number of layers was chosen as Deeper models showed proportionally increased hallucination rates [22]. Empirical analysis revealed that division by 12 normalizes the value relative to common transformer architectures.

4) Architecture Type Factor: A binary factor was implemented based on the model architecture:

$$f_{type} = \begin{cases} -0.07, & \text{if encoder-decoder} \\ -0.05, & \text{if encoder or decoder} \end{cases} \quad (15)$$

This accounts for the systematic differences observed between encoder-decoder and decoder-only architectures [23]. Supporting evidence shows that T5 and mT5 models had lower hallucination rates than similarly-sized GPT models. It was also observed that BART and other encoder-decoder architectures demonstrated better factual consistency.

5) Architectural Impact Calculation: Considering all the above factors, we arrived at the final Architectural Impact (A).

$$A = f_{layer} + f_{head} + f_{depth} + f_{type} \quad (16)$$

III. RESULTS AND ANALYSIS

A. Mathematical Model development

The final hallucination prediction formula integrates both the parameter-based and architectural components:

$$H_{total} = W + A \quad (17)$$

where:

$$W = (E_{Temp} \times 0.30) + (E_{Top_p} \times 0.25) + (E_{Top_k} \times 0.20) + (E_{Context_Window} \times 0.15) + (E_{Number_of_Tokens} \times 0.10) \quad (18)$$

$$A = f_{layer} + f_{head} + f_{depth} + f_{type} \quad (19)$$

Different from previous works which focus and quantify the hallucination level of LLMs using solely one parameter, this exploratory study provides empirical evidence that this new method can limit the model's hallucination within the desired range by fine tuning the various parameters. Our methodology showed that combinations of parameters and architecture of the models yield higher hallucination detection performance than any individual metric.This method also provided us a quantifiable contribution of each parameter on hallucination range and thereby help in mitigating hallucination.

1) Proof Generation: This section is devoted to the numerical simulations of the system in different cases to verify the obtained theoretical results. Three types of settings in the models were considered to get a thorough testing of the mathematical model

1) Standard setting strategy- It refers to the established and widely accepted approaches used for training, evaluating, and decoding from these models.
2) Greedy setting strategy- Greedy decoding generates text by selecting the token with the highest probability at each step of the sequence generation [24].
3) High Temperature setting strategy- It uses a larger temperature value, leading to more creative, diverse, and unpredictable outputs [25].

Case study 1: Encoder Model

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TABLE III
COMPARISON OF PREDICTED VS ACTUAL HALLUCINATION VALUES USING MATHEMATICAL MODEL OF ENCODER MODELS

Hallucination Rate Predictions and Errors (±10% Margin)						
Model	**Setting**	**Parameters**	**Pred.**	**Actual**	**Error**	**Within ±10%**
BERT Large	Standard	T=0.7, P=0.9, K=40, C=512, M=128	0.291	0.230	0.071	Yes
BERT Large	Greedy	T=0.0, P=1.0, K=1, C=512, M=128	0.286	0.215	0.071	Yes
BERT Large	High_temp	T=1.0, P=1.0, K=50, C=512, M=128	0.279	0.230	0.049	Yes
XLNet	Standard	T=0.7, P=0.9, K=40, C=512, M=128	0.291	0.218	0.073	Yes
XLNet	greedy	T=0.0, P=1.0, K=1, C=512, M=128	0.286	0.210	0.076	Yes
XLNet	High_temp	T=1.0, P=1.0, K=50, C=512, M=128	0.279	0.230	0.049	Yes

Case study 2: Decoder model

TABLE IV
COMPARISON OF PREDICTED VS ACTUAL HALLUCINATION VALUES USING THE MATHEMATICAL MODEL OF DECODER MODELS

Hallucination Rate Predictions and Errors (±10% Margin)						
Model	**Setting**	**Parameters**	**Pred.**	**Actual**	**Error**	**Within ±10%**
GPT-3	Standard	T=0.7, P=0.9, K=40, C=512, M=128	0.201	0.232	0.029	Yes
GPT-3	Greedy	T=0.0, P=1.0, K=1, C=512, M=128	0.198	0.224	0.026	Yes
GPT-3	High_temp	T=1.0, P=1.0, K=50, C=512, M=128	0.191	0.242	0.051	Yes
PaLM 8B	Standard	T=0.7, P=0.9, K=40, C=512, M=128	0.211	0.250	0.039	Yes
PaLM 8B	Greedy	T=0.0, P=1.0, K=1, C=512, M=128	0.206	0.230	0.024	Yes
PaLM 8B	High_temp	T=1.0, P=1.0, K=50, C=512, M=128	0.199	0.270	0.071	Yes

Case study 3: Encoder-Decoder model

TABLE V
COMPARISON OF PREDICTED VS ACTUAL HALLUCINATION VALUES USING THE MATHEMATICAL MODEL OF ENCODER-DECODER MODELS

Hallucination Rate Predictions and Errors (±10% Margin)						
Model	**Setting**	**Parameters**	**Pred.**	**Actual**	**Error**	**Within ±10%**
mT5 Base	Standard	T=0.7, P=0.9, K=40, C=512, M=128	0.293	0.210	0.083	Yes
mT5 Base	Greedy	T=0.0, P=1.0, K=1, C=512, M=128	0.288	0.205	0.083	Yes
mT5 Base	High_temp	T=1.0, P=1.0, K=50, C=512, M=128	0.281	0.225	0.056	Yes
T5 Large	Standard	T=0.7, P=0.9, K=40, C=512, M=128	0.271	0.195	0.076	Yes
T5 Large	Greedy	T=0.0, P=1.0, K=1, C=512, M=128	0.266	0.185	0.081	Yes
T5 Large	High_temp	T=1.0, P=1.0, K=50, C=512, M=128	0.259	0.205	0.054	Yes

The formula was validated against multiple language models including GPT-3, PaLM 8B, DistilBERT, XLNet, T5, GPT-4, BERT Large, and mT5 Base. All predictions fell within a ±10margin of error from actual hallucination rates, demonstrating the formula's reliability across different model architectures, sizes and parameters.

IV. CONCLUSION

Hallucination is the key hurdle to a LLM models credibility and poses a significant challenge in quantifying them. In this paper we developed a novel mathematical model that can predict the level of hallucination in various architectures of LLMs. We have provided a theoretical justification to our model. Different from previous works, this exploratory work provides empirical evidence that this new model can work on different LLM architectural models. All predictions fell within a ±10% margin of error from actual hallucination rates, demonstrating the formula's reliability across different model architectures, sizes and parameters. The significance of the work emerges from the fact that this work involves three realistic considerations. Overall, the proposed model is a contribution to the field of AI modeling and can serve as a valuable tool for everyone to be able to predict the percentage of hallucination. By understanding the key factors that drive the increase in hallucination, we can develop targeted and effective interventions to prevent and control it.

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